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Fundamental Concepts; a Simple One-Dimensional Boundary-Value Problem

1.1 INTRODUCTORY REMARKS AND PRELIMINARIES

The main constituents of a finite element method for the solution of a boundary-value problem are

- i. The variational or weak statement of the problem; and
- ii. The approximate solution of the variational equations through the use of "finite element functions."

To clarify concepts we shall begin with a simple example.

Suppose we want to solve the following differential equation for u :

$$u_{,xx} + \ell = 0 \quad (1.1.1)$$

where a comma stands for differentiation (i.e., $u_{,xx} = d^2u/dx^2$). We assume ℓ is a given smooth, scalar-valued function defined on the unit interval. We write

$$\ell : [0, 1] \rightarrow \mathbb{R} \quad (1.1.2)$$

where $[0, 1]$ stands for the unit interval (i.e., the set of points x such that $0 \leq x \leq 1$) and $\mathbb{R}$ stands for the real numbers. In words, (1.1.2) states that for a given x in $[0, 1]$, $\ell(x)$ is a real number. (Often we will use the notation $\in$ to mean "in" or "a member of." Thus for each $x \in [0, 1]$, $\ell(x) \in \mathbb{R}$.) Also, $[0, 1]$ is said to be the *domain* of ℓ , and $\mathbb{R}$ is its *range*.

We have described the given function ℓ as being smooth. Intuitively, you probably know what this means. Roughly speaking, if we sketch the graph of the function ℓ , we want it to be a smooth curve without discontinuities or kinks. We do this to avoid technical difficulties. Right now we do not wish to elaborate further as

this would divert us from the main theme. At some point prior to moving on to the next chapter, the reader may wish to consult Appendix 1.I, "An Elementary Discussion of Continuity, Differentiability and Smoothness," for further remarks on this important aspect of finite element work. The exercise in Sec. 1.16 already uses a little of the language described in Appendix 1.I. The terminology may be somewhat unfamiliar to engineering and physical science students, but it is now widely used in the finite element literature and therefore it is worthwhile to become accustomed to it.

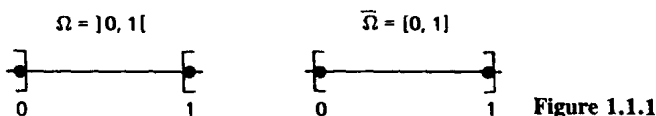
Equation (1.1.1) is known to govern the transverse displacement of a string in tension and also the longitudinal displacement of an elastic rod. In these cases, physical parameters, such as the magnitude of tension in the string, or elastic modulus in the case of the rod, appear in (1.1.1). We have omitted these parameters to simplify subsequent developments.

Before going on, we introduce a few additional notations and terminologies. Let $]0, 1[$ denote the unit interval without end points (i.e., the set of points x such that $0 < x < 1$). $]0, 1[$ and $[0, 1]$ are referred to as *open and closed unit intervals*, respectively. To simplify subsequent writing and tie in with notation employed later on in multidimensional situations, we shall adopt the definitions

$$\Omega =]0, 1[\quad (\text{open}) \quad (1.1.3)$$

$$\bar{\Omega} = [0, 1] \quad (\text{closed}) \quad (1.1.4)$$

See Fig. 1.1.1.



At this point, considerations such as these may seem pedantic. Our purpose, however, is to develop a language for the precise articulation of boundary-value problems, which is necessary for good finite element work.

1.2 STRONG, OR CLASSICAL, FORM OF THE PROBLEM

A boundary-value problem for (1.1.1) involves imposing *boundary conditions* on the function u . There are a variety of possibilities. We shall assume u is required to satisfy

$$u(1) = q \quad (1.2.1)$$

$$-u_{,x}(0) = h \quad (1.2.2)$$

where q and h are given constants. Equations (1.2.1) and (1.2.2) require that u take on the value q at $x = 1$ and the derivative of u (i.e., slope) take on the value $-h$ at $x = 0$, respectively. This set of boundary conditions will later enable us to illustrate certain key features of variational formulations. For obvious reasons, boundary conditions of the type (1.2.1) and (1.2.2) lead to so-called *two-point boundary-value problems*.

The strong form of the boundary-value problem, (S) , is stated as follows:

$$(S) \left\{ \begin{array}{l} \text{Given } \ell : \bar{\Omega} \rightarrow \mathbb{R} \text{ and constants } q \text{ and } h, \text{ find } u : \bar{\Omega} \rightarrow \mathbb{R}, \text{ such that} \\ u_{,xx} + \ell = 0 \quad \text{on } \Omega \\ u(1) = q \\ -u_{,x}(0) = h \end{array} \right.$$

When we write $u_{,xx} + \ell = 0$ on Ω we mean $u_{,xx}(x) + \ell(x) = 0$ for all $x \in \Omega$. Of course, the exact solution of (S) is trivial to obtain, namely,

$$u(x) = q + (1 - x)h + \int_x^1 \left\{ \int_0^y \ell(z) dz \right\} dy \quad (1.2.3)$$

where y and z are used to denote dummy variables. However, this is not the main concern here. We are interested in developing schemes for obtaining approximate solutions to (S) that will be applicable to much more complex situations in which exact solutions are not possible.

Some methods of approximation begin directly with the strong statement of the problem. The most notable example is the finite difference method (e.g., see [1]). The finite element method requires a different formulation, which is treated in the next section.

1.3 WEAK, OR VARIATIONAL, FORM OF THE PROBLEM

To define the weak, or variational, counterpart of (S) , we need to characterize two classes of functions. The first is to be composed of candidate, or trial, solutions. From the outset, we shall require these possible solutions to satisfy the boundary condition $u(1) = q$. The other boundary condition will not be required in the definition. Furthermore, so that certain expressions to be employed make sense, we shall require that the derivatives of the trial solutions be square-integrable. That is, if u is a trial solution, then

$$\int_0^1 (u_{,x})^2 dx < \infty \quad (1.3.1)$$

Functions that satisfy (1.3.1) are called H^1 -functions; we write $u \in H^1$. Sometimes the domain is explicitly included, i.e., $u \in H^1([0, 1])$.

Thus the collection of *trial solutions*, denoted by $\mathcal{S}$, consists of all functions which have square-integrable derivatives and take on the value q at $x = 1$. This is written as follows:

$$\mathcal{S} = \{u \mid u \in H^1, u(1) = q\} \quad (\text{trial solutions}) \quad (1.3.2)$$

The fact that $\mathcal{S}$ is a collection, or set, of objects is indicated by the curly brackets (called braces) in (1.3.2). The notation for the typical member of the set, in this case u , comes first inside the left-hand curly bracket. Following the vertical line ($|$) are the properties satisfied by members of the set.

The second collection of functions is called the *weighting functions*, or *variations*. This collection is very similar to the trial solutions except we require the homogeneous counterpart of the q -boundary condition. That is, we require weighting functions, w , to satisfy $w(1) = 0$. The collection is denoted by $\mathcal{U}$ and defined by

$$\mathcal{U} = \{w \mid w \in H^1, w(1) = 0\} \quad (\text{weighting functions}) \quad (1.3.3)$$

It simplifies matters somewhat to continue to think of $f: \Omega \rightarrow \mathbb{R}$ as being smooth. (However, what follows holds for a considerably larger class of f 's.)

In terms of the preceding definitions, we may now state a suitable weak form, (W) , of the boundary-value problem.

$$(W) \left\{ \begin{array}{l} \text{Given } f, q, \text{ and } h, \text{ as before. Find } u \in \mathcal{S} \text{ such that for all } w \in \mathcal{U} \\ \hline \int_0^1 w_{,x} u_{,x} dx = \int_0^1 w f dx + w(0)h \end{array} \right. \quad (1.3.4)$$

Formulations of this type are often called *virtual work*, or *virtual displacement, principles* in mechanics. The w 's are the *virtual displacements*.

Equation (1.3.4) is called the *variational equation*, or (especially in mechanics) the *equation of virtual work*.

The solution of (W) is called the *weak, or generalized, solution*. The definition given of a weak formulation is not the only one possible, but it is the most natural one for the problems we wish to consider.

1.4 EQUIVALENCE OF STRONG AND WEAK FORMS; NATURAL BOUNDARY CONDITIONS

Clearly, there must be some relationship between the strong and weak versions of the problem, or else there would be no point in introducing the weak form. It turns out that the weak and strong solutions are identical. We shall establish this assuming all functions are smooth. This will allow us to proceed expeditiously without invoking technical conditions with which the reader is assumed to be unfamiliar. "Proofs" of this kind are sometimes euphemistically referred to as "formal proofs." The intent is not to be completely rigorous but rather to make plausible the truth of the proposition. With this philosophy in mind, we shall "prove" the following.

Proposition

- a. Let u be a solution of (S) . Then u is also a solution of (W) .
- b. Let u be a solution of (W) . Then u is also a solution of (S) .

Another result, which we shall not bother to verify but is in fact easily established,

is that both (S) and (W) possess unique solutions. Thus, by (a) and (b), the strong and weak solutions are one and the same. Consequently, (W) is equivalent to (S) .

Formal Proof

a. Since u is assumed to be a solution of (S) , we may write

$$0 = - \int_0^1 w(u_{,xx} + f) dx \quad (1.4.1)$$

for any $w \in \mathcal{U}$. Integrating (1.4.1) by parts results in

$$0 = \int_0^1 w_{,x} u_{,x} dx - \int_0^1 w f dx - w u_{,x} \Big|_0^1 \quad (1.4.2)$$

Rearranging and making use of the fact that $-u_{,x}(0) = h$ and $w(1) = 0$ results in

$$\int_0^1 w_{,x} u_{,x} dx = \int_0^1 w f dx + w(0)h \quad (1.4.3)$$

Furthermore, since u is a solution of (S) , it satisfies $u(1) = q$ and therefore is in $\mathcal{S}$. Finally, since u also satisfies (1.4.3) for all $w \in \mathcal{U}$, u satisfies the definition of a weak solution given by (W) .

b. Now u is assumed to be a weak solution. Thus $u \in \mathcal{S}$; consequently $u(1) = q$, and

$$\int_0^1 w_{,x} u_{,x} dx = \int_0^1 w f dx + w(0)h$$

for all $w \in \mathcal{U}$. Integrating by parts and making use of the fact $w(1) = 0$ results in

$$0 = \int_0^1 w(u_{,xx} + f) dx + w(0)[u_{,x}(0) + h] \quad (1.4.4)$$

To prove u is a solution of (S) it suffices to show that (1.4.4) implies¹

- i. $u_{,xx} + f = 0$ on Ω ; and
- ii. $u_{,x}(0) + h = 0$

First we shall prove (i). Define w in (1.4.4) by

$$w = \phi(u_{,xx} + f) \quad (1.4.5)$$

where ϕ is smooth; $\phi(x) > 0$ for all $x \in \Omega =]0, 1[$; and $\phi(0) = \phi(1) = 0$. For example, we can take $\phi(x) = x(1 - x)$, which satisfies all the stipulated requirements (see Figure 1.4.1). It follows that $w(1) = 0$ and thus $w \in \mathcal{U}$, so (1.4.5) defines a

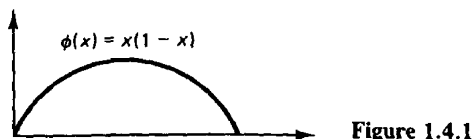


Figure 1.4.1

¹These equations are sometimes called the *Euler-Lagrange equations* of the weak formulation.

legitimate member of $\mathcal{U}$. Substituting (1.4.5) into (1.4.4) results in

$$0 = \int_0^1 \underbrace{\phi(u_{,xx} + f)^2}_{\geq 0} dx + 0 \quad (1.4.6)$$

Since $\phi > 0$ on Ω , it follows from (1.4.6) that (i) must be satisfied.

Now that we have established (i), we may use it in (1.4.4) to prove (ii), namely,

$$0 = w(0)[u_{,x}(0) + h] \quad (1.4.7)$$

That $w \in \mathcal{U}$ puts no restriction whatsoever on its value at $x = 0$. Therefore, we may assume that the w in (1.4.7) is such that $w(0) \neq 0$. Thus (ii) is also shown to hold, which completes the proof of the proposition. ■

Remarks

1. The boundary condition $-u_{,x}(0) = h$ is not explicitly mentioned in the statement of (W). From the preceding proof, we see that this boundary condition is, however, implied by the satisfaction of the variational equation. Boundary conditions of this type are referred to as *natural boundary conditions*. On the other hand, trial solutions are explicitly required to satisfy the boundary condition $u(1) = q$. Boundary conditions of this type are called *essential boundary conditions*. The fact that solutions of the variational equation satisfy natural boundary conditions is extremely important in more complicated situations which we will consider later on.

2. The method used to prove part (b) of the proposition goes under the name of the *fundamental lemma* in the literature of the calculus of variations. In essence, it is the methodology that enables us to deduce the differential equations and boundary conditions implied by the weak formulation. To develop correct weak forms for complex, multidimensional problems, it is essential to have a thorough understanding of these procedures.

Now we see that to obtain approximate solutions to the original boundary-value problem we have alternative starting points, i.e., the strong or weak statements of the problem. *Finite element methods are based upon the latter*. Roughly speaking, the basic idea is to approximate $\mathcal{S}$ and $\mathcal{U}$ by convenient, finite-dimensional collections of functions. (Clearly, $\mathcal{S}$ and $\mathcal{U}$ contain infinitely many functions.) The variational equations are then solved in this finite-dimensional context. An explicit example of how to go about this is the subject of the next section. However, we first introduce some additional notations to simplify subsequent writing.

Let

$$a(w, u) = \int_0^1 w_{,x} u_{,x} dx \quad (1.4.8)$$

$$(w, f) = \int_0^1 w f dx \quad (1.4.9)$$